A continuum of Schatten counterexamples with normal off-diagonal block

Partial answer to arXiv:2111.15180

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Status. New proved partial result. The proposed inequality fails for every $p > q_* = 7.709989669514\ldots$. The remaining finite-exponent range is not classified here.

1 Question and result

Bourin and Lee ask, after Proposition 3.4 of [1], for which $p \in [1, \infty]$ the inequality

$$\left\| \begin{bmatrix} A & N \\ N^* & B \end{bmatrix} \right\|_p \leq \|A + B\|_p \tag{1}$$

holds for every positive block matrix when N is normal. Their proposition proves (1) for $p = 2$; for $p = 1$ both sides equal $\operatorname{Tr}(A + B)$.

Define $q_* > 1$ to be the second positive solution of

$$\left(\frac{35}{32}\right)^q + 2\left(\frac{29}{64}\right)^q = 2. \tag{2}$$

Numerically,

$$q_* = 7.709989669514\ldots.$$

Theorem 1. *For every $p > q_*$, including $p = \infty$, there are $A, B \in M_3^+$ and a unitary $N \in M_3$ such that $\left[\begin{smallmatrix} A & N \\ N^* & B \end{smallmatrix}\right] \geq 0$ but (1) is false.*

Thus the set of exponents sought in [1] is contained in $[1, q_*]$. The theorem strengthens a single operator-norm counterexample to an explicit half-line of Schatten exponents.

2 Exact construction

Put

$$A = \operatorname{diag}\left(\frac{2}{5}, \frac{5}{2}, 6\right), \qquad C = \operatorname{diag}\left(\frac{5}{2}, \frac{2}{5}, 6\right), \qquad N = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix},$$

and set

$$B = N^* C N = \operatorname{diag}\left(6, \frac{5}{2}, \frac{2}{5}\right), \qquad H = \begin{bmatrix} A & N \\ N^* & B \end{bmatrix}.$$

Certainly N is unitary, hence normal. Congruence by $\operatorname{diag}(I, N^*)$ gives

$$H \simeq \begin{bmatrix} A & I \\ I & C \end{bmatrix}.$$

Because A and C are diagonal, the latter matrix is permutation-similar to the direct sum

$$\begin{bmatrix} 2/5 & 1 \\ 1 & 5/2 \end{bmatrix} \oplus \begin{bmatrix} 5/2 & 1 \\ 1 & 2/5 \end{bmatrix} \oplus \begin{bmatrix} 6 & 1 \\ 1 & 6 \end{bmatrix}.$$

The first two summands have determinant zero and the last is positive. Consequently $H \geq 0$, and its eigenvalues are exactly

$$7,\ 5,\ \frac{29}{10},\ \frac{29}{10},\ 0,\ 0. \tag{3}$$

On the other hand,

$$A + B = \operatorname{diag}\left(\frac{32}{5}, 5, \frac{32}{5}\right). \tag{4}$$

For finite $p \geq 1$, equations (3)–(4) show that $\|H\|_p > \|A + B\|_p$ precisely when

$$7^p + 5^p + 2(29/10)^p > 2(32/5)^p + 5^p,$$

or, after cancellation and normalization, precisely when

$$g(p) := \left(\frac{35}{32}\right)^p + 2\left(\frac{29}{64}\right)^p > 2. \tag{5}$$

It remains only to justify the stated threshold. We have $g(1) = 2$ and $g''(p) > 0$. Moreover $g'(1) < 0$: indeed

$$35 \log(35/32) + 29 \log(29/64) < 35\frac{3}{32} - 29\frac{1}{2} < 0,$$

using $\log(1 + x) < x$ and $64/29 > 2$ with $\log 2 > 1/2$. Finally $g(p) \to \infty$, since $35/32 > 1$. Strict convexity therefore gives exactly one further intersection $q_* > 1$ with the level 2, and $g(p) > 2$ exactly for $p > q_*$. This proves the finite-p assertion. At $p = \infty$, (3)–(4) give $\|H\|_\infty = 7 > 32/5 = \|A+B\|_\infty$.

3 Provenance and remaining gap

Hayashi's Problem 3 and its following example [2, PDF pp. 5–6] use the same cyclic-unitary reduction, with different diagonal entries, to disprove the operator-norm inequality. The rational choice above is tuned so that one eigenvalue cancels and the entire Schatten comparison becomes the two-exponential equation (2).

No claim is made for $1 < p < 2$ or $2 < p \leq q_*$. Endpoint interpolation does not by itself compare the Schatten norms of two different matrices, and tensoring or taking direct sums preserves the sign of a fixed p-norm comparison. These are substantive obstructions to upgrading this packet to a full characterization.

References

[1] J.-C. Bourin and E.-Y. Lee, *Eigenvalue inequalities for positive block matrices with the inradius of the numerical range*, Int. J. Math. 33 (2022), 2250009; arXiv:2111.15180, Proposition 3.4 and the question on PDF page 7.

[2] T. Hayashi, *On a norm inequality for a positive block-matrix*, Linear Algebra Appl. 566 (2019), 86–97; arXiv:1808.00181.